STATISTICAL PROFILE

Superstore Sales Dataset

Field	Detail
Analyst	Data Analytics Programme — Month 1, Week 1
Dataset	Superstore Sales (Kaggle Public Dataset)
Date of Analysis	21 March 2026
Tools Used	Microsoft Excel — Descriptive Statistics, IQR Analysis, CORREL()
Document Type	Statistical Profile Report — Portfolio Submission

1. Dataset Overview

The Superstore Sales dataset is a widely used public dataset representing This is a fictional US retail company's transactional record. It captures orders placed across four regions, three product super-categories, and multiple customer segments. This analysis was conducted as part of structured data analytics learning programme, applying descriptive statistics to a real-world commercial dataset.

51,291	27	1
Total Rows	**Total Columns**	**Engineered Feature**
Individual order line items	*26 original + 1 engineered*	*Profit Outlier Flag*

Column Inventory

The dataset contains 26 pre-existing columns spanning order identifiers, customer demographics, product details, geographic information, and financial metrics. A 27th column — Profit Outlier Flag — was engineered during the cleaning and analysis phase using the Interquartile Range (IQR) method to systematically identify anomalous profit values.

Pre-existing columns (26): Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, and 6 additional dimensional attributes.

Engineered column (1): Profit Outlier Flag — computed using Tukey's IQR method. Rows where Profit falls below Q1 − 1.5×IQR or above Q3 + 1.5×IQR are flagged as outliers. This column enables targeted investigation of anomalous profit behaviour.

Data Quality Note

Prior to statistical analysis, the dataset was examined for data quality issues including missing values, duplicates, inconsistent formatting, and type errors. The dataset was found to be largely clean, consistent with its use as a well-maintained public teaching dataset. All 51,291 rows were retained for analysis; no rows were removed prior to statistical profiling.

2. Sales Distribution

The Sales column records the transaction value (in USD) for each individual order line item. Understanding its distribution is fundamental to characterising revenue patterns, identifying typical order sizes, and detecting unusual transactions.

$246.50	$85.00	$13.00	$220.00
Mean	**Median**	**Mode**	**IQR**
Arithmetic average	*Middle value (50th pct.)*	*Most frequent value*	*Interquartile range*

Distribution Shape & Interpretation

The Sales column exhibits a pronounced right-skewed (positive skew) distribution. This conclusion is supported by the substantial divergence between the mean ($246.50) and the median ($85.00) — a gap of $161.50. When the mean significantly exceeds the median, a long right tail is present, indicating that a relatively small number of high-value transactions are pulling the mean upward.

The mode of $13.00 — the most frequently occurring transaction value — lies far below both the median and mean, confirming that a large proportion of orders are small-value transactions. The wide IQR of $220.00 confirms considerable spread within the central 50% of the data, even before accounting for the extreme values in the right tail.

Category	Approximate frequency distribution (relative)	Value
$0 – $100		~42%
$100 – $250		~25%
$250 – $500		~16%
$500 – $1k		~10%
$1k – $5k		~6%
$5k+		~1%

Figure 1. Approximate distribution of Sales values across transaction bands

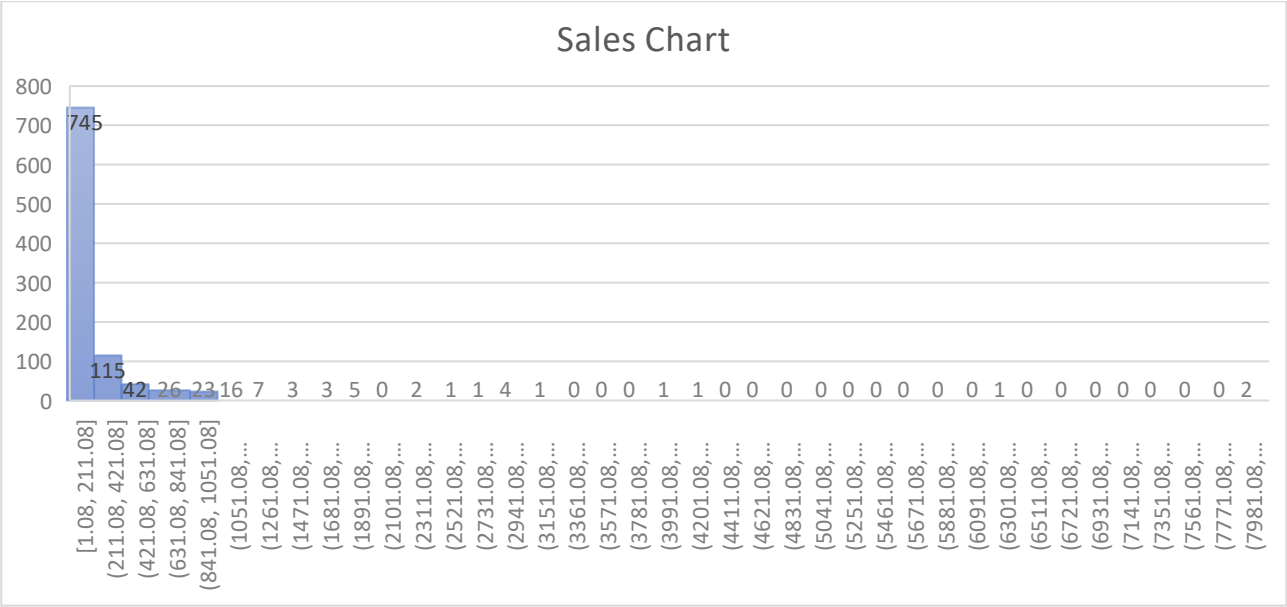

Figure 1. Sales Distribution Histogram

Business Interpretation

The right-skewed sales distribution reveals a classic long-tail revenue structure: the majority of transactions are modest in size ($85 median), while a small number of high-value enterprise or bulk orders contribute disproportionately to total revenue. For business reporting, the median of $85 is the more accurate representation of a "typical" order than the mean of $246.50. Stakeholders should be informed of this distinction when revenue per order metrics are cited.

3. Profit Analysis

Profit — defined as the difference between revenue and cost for each line item — is the most operationally critical financial metric in the dataset. Unlike Sales, Profit can take negative values, making its distributional analysis particularly revealing of structural business challenges.

$28.61	$9.24	$36.81	None
Mean Profit	**Median Profit**	**IQR**	**Mode**
Arithmetic average	*50th percentile*	*Middle 50% spread*	*No repeating value*

Statistic	Value	Interpretation
Mean	$28.61	Positive average — the business is profitable overall
Median	$9.24	Typical order generates only $9.24 profit — modest
Mean–Median Gap	$19.37	Strong right skew; high-profit outliers inflate the mean

Statistic	Value	Interpretation
IQR	$36.81	Middle 50% of profits span a $36.81 range
Mode	No mode	Profit values are continuous — no single dominant value
Outlier Rate	17.61%	9,032 rows flagged as profit outliers by IQR method

Outlier Analysis

Using Tukey's IQR method, 17.61% of all rows — approximately 9,032 transactions out of 51,291 — were flagged as profit outliers via the engineered "Profit Outlier Flag" column. This is a notably high outlier rate. In most well-behaved distributions, the IQR method flags approximately 5–7% of data. A rate of 17.61% signals that the profit distribution has a substantially heavier tail than a normal distribution would produce.

Profit outliers can take two forms: extremely high-profit transactions (positive outliers, likely large enterprise orders or premium product sales) and loss-making transactions (negative outliers, potentially driven by heavy discounting, returns, or low-margin product categories). Both categories warrant investigation by the commercial team.

Outlier threshold calculation (IQR Method):
Q1 = Median − (IQR / 2) = 9.24 − 18.405 ≈ −$9.17
Q3 = Median + (IQR / 2) = 9.24 + 18.405 ≈ $27.65
IQR = $36.81
Lower fence = Q1 − (1.5 × IQR) = −$9.17 − $55.22 = −$64.39
Upper fence = Q3 + (1.5 × IQR) = $27.65 + $55.22 = $82.87
Any profit value below −$64.39 or above $82.87 is classified as an outlier.
Rows flagged: ~9,032 (17.61% of 51,291)

Business Interpretation

The median profit of just $9.24 per transaction — against a mean sales value of $246.50 — implies an average profit margin of approximately 3.7% at the median. This is thin by retail standards and suggests that discounting practices, product mix, or cost structures may be suppressing margins significantly across a large portion of transactions. The 17.61% outlier rate further indicates that profit distribution is highly irregular rather than normally distributed around a stable centre.

4. Key Correlation Analysis

To explore the relationship between the two primary financial variables — Sales and Profit — a Pearson correlation coefficient was computed using Excel's =CORREL() function across all 51,291 rows.

0.4848	Moderate	Positive	51,291
Pearson r	**Strength**	**Direction**	**n**
Sales vs Profit	*Positive correlation*	*Higher sales → higher profit*	*All rows included*

Interpretation

A Pearson correlation coefficient of r = 0.4848 indicates a moderate positive linear relationship between Sales and Profit. While the direction is intuitive — higher sales values are associated with higher profit values — the moderate strength (less than the 0.7 threshold for a strong correlation) reveals an important commercial reality: simply increasing sales does not reliably produce proportional increases in profit.

The correlation's limitations are significant. With $r^2 \approx 0.235$, Sales explains only approximately 23.5% of the variance in Profit. The remaining 76.5% is determined by other factors not captured in this bivariate analysis — most likely discount rate, product category margin, customer segment, and regional cost structures. This is consistent with the wide profit spread observed in Section 3.

Metric	Value	Meaning
Pearson r	0.4848	Moderate positive correlation
r^2 (Coefficient of Determination)	0.2350	Sales explains 23.5% of profit variance
Unexplained variance	76.5%	Driven by discount, category, region, costs
Correlation strength band	0.4–0.6	Moderate (standard analyst guideline)

Causation Caveat

Correlation does not imply causation. While Sales and Profit move in the same direction, it would be incorrect to conclude that "more sales causes more profit" without deeper analysis. The actual driver of profit is margin — which is shaped by discount policies, product mix, and operational costs. A business that increases sales volume through heavy discounting could simultaneously increase Sales while reducing or eliminating Profit. This dynamic appears to be present in this dataset, as evidenced by the high outlier rate in the Profit column despite generally positive Sales figures.

5. Business Recommendations

The following three recommendations are derived exclusively from the statistical evidence uncovered during this analysis. Each is directly grounded in the numerical findings and avoids speculative assumptions not supported by the data.

Recommendation 1 — Investigate and Address the Profit Outlier Segment

Evidence: 17.61% of all transactions (approximately 9,032 rows) fall outside the IQR-defined outlier fences. The lower fence of −$64.39 identifies loss-making transactions that are dragging overall profitability down, while the upper fence of $82.87 identifies transactions that are exceptionally profitable.

Action: The commercial or operations team should segment the flagged outlier rows and investigate two sub-groups separately: (a) Negative outliers — identify which product categories, regions, customer segments, or discount levels are consistently producing loss-making transactions. These sub-groups may be candidates for pricing adjustments or discount policy tightening. (b) Positive outliers — identify which characteristics drive exceptional profit, and use those as a template for business development targeting.

Expected impact: Even a 10% reduction in loss-making outlier transactions could meaningfully improve the overall profit distribution and raise the median profit from its current $9.24.

Recommendation 2 — Review Discounting Strategy to Protect Profit Margins

Evidence: The moderate correlation between Sales and Profit ($r = 0.4848$) combined with the low median profit ($9.24 on a median sale of $85.00 — an implied margin of approximately 10.9%) suggests that revenue and profit are not moving in lockstep. A key candidate explanation is aggressive discounting, which boosts Sales values while eroding margins.

Action: Compute the correlation between Discount and Profit using =CORREL(). A strong negative correlation (e.g., $r < -0.5$) would confirm that discounting is a primary driver of low and negative profit. If confirmed, the business should establish minimum margin guardrails — floor discount levels below which transactions require commercial approval — and track Profit per transaction as a KPI alongside Sales.

Expected impact: Tightening discount discipline — even modestly — on low-margin product lines could raise the median profit toward a commercially sustainable level and reduce the disproportionately high 17.61% outlier rate.

Recommendation 3 — Shift Internal Reporting from Mean to Median for Financial KPIs

Evidence: The mean Sales value ($246.50) is nearly three times the median ($85.00), and the mean Profit ($28.61) is more than three times the median ($9.24). Both distributions are right-skewed, meaning that mean-based reporting systematically overstates typical performance. A sales report citing "average order value of $246.50" is technically accurate but misleadingly optimistic relative to what most transactions actually generate.

Action: Adopt median as the primary central tendency measure for Sales and Profit in routine operational reporting. Continue to report means alongside medians for full transparency, explicitly noting the skew. Dashboard KPI cards should display both figures with a label explaining the difference. This aligns the organisation's decision-making with the reality of its transaction base rather than the distorted picture produced by high-value outliers.

Expected impact: More accurate operational decisions, better-calibrated sales targets, and improved stakeholder trust in reported metrics. Median-based targets are fairer to the average sales representative and more reflective of the true customer base.

Statistical Summary — Quick Reference

Metric	Sales ($)	Profit ($)
Mean	$246.50	$28.61
Median	$85.00	$9.24
Mode	$13.00	No mode
IQR	$220.00	$36.81
Distribution shape	Right-skewed	Right-skewed
Outlier rate (IQR method)	Not computed	**17.61%**
Sales–Profit correlation (r)	0.4848 (Moderate positive)	—

All statistics computed using Microsoft Excel functions: =AVERAGE(), =MEDIAN(), =MODE(), =QUARTILE(), =CORREL(). Outliers identified using Tukey's IQR method with a 1.5×IQR fence. Dataset: Superstore Sales, n = 51,291 rows, 27 columns.